

Fragility of Lasso Variable Selection:

Removal Witnesses, Exact Deletion Tests, Non-Monotonicity and Computational Status

Luciano Silva Alarco

Pontificia Universidad Católica del Perú

ORCID 0000-0002-3395-3129

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Abstract

We study how the set of variables selected by the Lasso changes when observations are removed. For data D , a penalty μ and a subsample penalty rule, a *removal witness* for a target change (a variable leaves, one enters, or the selected set changes) is a set R of observations whose removal produces the change; the *fragility number* is the minimal $|R|$. We establish: (i) an exact closed-form test of whether removing R preserves the signed support, which for $|R| = 1$ under a constant penalty is a Cook–Belsley deletion formula with the Lasso residual, plus a cheap single-removal certificate and a loose polynomial one for k removals; (ii) non-monotonicity of witness families, by hand for $p = 1$ and in exact arithmetic for $p = 2$; (iii) weak NP-completeness of deselection-witness existence already for $p = 1$, where selection is solvable in $O(n \log n)$, and of selection-witness existence for every fixed $p \geq 2$, with a pseudo-polynomial algorithm for every fixed p ; (iv) on two synthetic Gaussian families with $n \leq 14$, exact fragility numbers by exhaustive search (one removal changes the selection in 92%/52% of the instances) and three heuristics failing a criterion fixed in advance, since at $n \leq 14$ none is cheaper than one tenth of the exhaustive search. No novelty is claimed for the algebra.

1 Introduction

Variable selection with the Lasso [21] is known to be unstable: small changes of the data can change the selected set. This instability has been addressed statistically, by aggregating selections over random subsamples [stability selection, 15], and asymptotically, by conditions under which the selected set is the true one with high probability [5, 23, 25]. A different, finite-sample and combinatorial question is the subject of the research line to which this draft belongs (TCD001): *how many observations must be removed, and which, for the selected set to change?* The question has a long history for the coefficients of least squares, where case-deletion diagnostics [3, 6] and the influence curve [11] measure the effect of dropping one observation, and where the worst-case effect of dropping a small fraction of the data has recently been studied under the name of approximate maximum influence perturbation [4], through adaptive heuristics for influential sets [14], through exact or certified algorithms for the sign of a regression coefficient [8, 16, 19], and, for estimands whose leave-set-out effect is linear-fractional, through a reduction of the most influential set to a sequence of top- k problems [13]. For a *selection*, the quantity of interest is discrete, so the influence function does not apply directly, and the natural object is a set of observations, a *witness*, together with its minimal size.

The public record of the line states that non-monotone changes and connections to known combinatorial problems were identified, and that novelty, computational advantage and some certificates were not sufficiently established. This draft gives the line a precise core: it proves what can be proved, reports which certificates are loose, and shows that on small synthetic instances the predefined advantage criterion is not met because exhaustive search with the closed-form test is already cheap; novelty is not claimed. Section 2 fixes the definitions; Section 3 proves the exact deletion test (Proposition 3.1), a Cook–Belsley deletion formula with the Lasso residual (Corollary 3.2) and a polynomial certificate that no k removals change the support (Proposition 3.4); Section 4 shows that witness families are not closed under supersets ($p = 1$ by hand, $p = 2$ in exact arithmetic, by competition between variables); in Section 5, existence of a deselection witness is NP-complete for $p = 1$ by a reduction from SUBSET SUM and selection witnesses for $p = 1$ are easy, whereas from $p = 2$ on selection witnesses are NP-complete too (Theorem 5.4: a competing variable absorbs the target unless a subset sum is exact); a pseudo-polynomial algorithm for every fixed p (Proposition 5.5) makes all this hardness weak, and only strong hardness with unbounded p remains open (Conjecture 5.8); Section 6

reports exact fragility numbers, non-monotone pairs, scaling with n , and three heuristics against the exact minimum under a criterion fixed before the runs. The paper is methodological: it makes no claim about real data and does not propose the fragility number as a replacement for stability selection (Remark 3.5 states the one implication that follows from the definitions); every number in the text is produced by the scripts of Appendix A with fixed seeds, and Section 7 lists all claims with their status.

2 Setting and definitions

Let $D = (X, y)$ with $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$, no intercept and columns as given. For $\mu > 0$ the Lasso objective in sum form is

$$L_D(\beta; \mu) = \frac{1}{2} \|y - X\beta\|_2^2 + \mu \|\beta\|_1, \quad (1)$$

which is the objective of scikit-learn’s **Lasso** with **alpha** = μ/n [18]. We write $\hat{\beta}(D; \mu)$ for its minimiser when it is unique, $S(D) = \text{supp } \hat{\beta}$ for the *support* (the selected set) and (S, s) with $s = \text{sgn } \hat{\beta}_S \in \{\pm 1\}^S$ for the *signed support*. Two facts from Tibshirani [22] are used throughout: if the columns of X are in general position the minimiser is unique for every y and $\mu > 0$; and if the *equicorrelation set* $E = \{j : |X_j^\top (y - X\hat{\beta})| = \mu\}$ has X_E of full column rank, the minimiser is unique and determined by the KKT system on E . Conversely a unique minimiser has X_S of full column rank (Lemma 2.4). [literature]

Definition 2.1 (Penalty rule). A *penalty rule* assigns to the subsample $D \setminus R$ of size $n - k$ a penalty μ_k . Two rules are used: the *constant rule* (C), $\mu_k = \mu$, and the *proportional rule* (P), $\mu_k = \mu(n - k)/n$, which keeps scikit-learn’s **alpha** fixed and is what a practitioner does who refits “with the same λ ” in the mean-squared-error parametrisation.

Definition 2.2 (Removal witness, fragility number). Fix D , μ , a rule, and let (S, s) be the signed support on D . A *target* Π is a property of the signed support of the Lasso on a subsample, relative to (S, s) ; the targets used here are

- ANY: $S(D \setminus R) \neq S$ (unsigned), or $(S(D \setminus R), s(D \setminus R)) \neq (S, s)$ (signed);
- LEAVE(j), $j \in S$: $j \notin S(D \setminus R)$;
- ENTER(j), $j \notin S$: $j \in S(D \setminus R)$.

A *removal witness* for Π is a nonempty proper subset $\emptyset \neq R \subsetneq [n]$ such that the Lasso on $D \setminus R$ with penalty $\mu_{|R|}$ has a unique minimiser and Π holds. The *witness family* is $W_\Pi(D) = \{R : R \text{ is a witness}\}$ and the *fragility number* is $f_\Pi(D) = \min\{|R| : R \in W_\Pi(D)\} \geq 1$, with $f_\Pi(D) = \infty$ if $W_\Pi(D) = \emptyset$.

The signed and unsigned versions of ANY differ only when a coefficient changes sign without leaving the support; $f_{\text{ANY}}^{\text{signed}} \leq f_{\text{ANY}}$ and the experiments report how often the two differ. Witnesses for the signed target are decided without refitting by Proposition 3.1.

Definition 2.3 (Monotonicity). The target Π is *monotone on D* if $R \in W_\Pi(D)$ and $R \subseteq R' \subsetneq [n]$ imply $R' \in W_\Pi(D)$. A *non-monotone pair* is a pair $R \subsetneq R'$ with $R \in W_\Pi(D)$ and $R' \notin W_\Pi(D)$; for LEAVE(j) this means that j leaves the support after removing R and re-enters after removing R' .

Lemma 2.4. [classical] *If the minimiser of (1) is unique with support S , then X_S has full column rank.*

Proof. If $X_S \eta = 0$ with $\eta \neq 0$, extend η by zeros. For $|t|$ small, $\hat{\beta} + t\eta$ has the same fit and the same signs on S , so $L_D(\hat{\beta} + t\eta) = L_D(\hat{\beta}) + t\mu s^\top \eta$. If $s^\top \eta \neq 0$ one sign of t decreases the objective, contradicting optimality; if $s^\top \eta = 0$ every such $\hat{\beta} + t\eta$ is a minimiser, contradicting uniqueness. (This is part of the characterisation in 22.) $\square$

3 An exact deletion test and two certificates

Throughout this section $\hat{\beta}$ is the unique minimiser on D with signed support (S, s) , $|S| = s_0 \geq 0$ (we write s_0 for the size to avoid a clash with the sign vector), X_S has full column rank, and

$$\begin{aligned} A &= X_S^\top X_S, & r &= y - X\hat{\beta}, & c &= X^\top r, & V &= X_S A^{-1} \text{ (rows } v_i^\top), \\ h_i &= x_{iS}^\top A^{-1} x_{iS}, & \tilde{X} &= (I - X_S A^{-1} X_S^\top) X_S c, \end{aligned}$$

so that $c_S = \mu s$, $|c_j| \leq \mu$ for $j \notin S$, $h_i \in [0, 1]$ are the leverages of the active-set regression and $\tilde{X}_j$ is the residual of X_j regressed on X_S . For $R \subseteq [n]$ with $|R| = k$ let $U = X_{R,S} \in \mathbb{R}^{k \times s_0}$ and $H_R = U A^{-1} U^\top$, the $R \times R$ block of the projection $X_S A^{-1} X_S^\top$.

Proposition 3.1 (Exact removal test). [proved here] Let $R \subsetneq [n]$, $|R| = k$, and let μ' be the penalty on $D \setminus R$, $\delta = \mu - \mu'$.

(a) If $I - H_R$ is singular, (S, s) is not the signed support of a unique minimiser on $D \setminus R$.

(b) If $I - H_R$ is nonsingular, put $e_R = (I - H_R)^{-1}r_R$, $w = A^{-1}s$, $z = w + V_R^\top(I - H_R)^{-1}Uw$, and

$$\tilde{\beta}_S = \hat{\beta}_S - V_R^\top e_R + \delta z, \quad (2)$$

$$\tilde{c}_j = c_j - \tilde{X}_{R,j}^\top e_R - \delta(X_j^\top X_S - X_{R,j}^\top U)z, \quad j \notin S. \quad (3)$$

Then (S, s) is the signed support of a minimiser on $D \setminus R$ if and only if $s_j \tilde{\beta}_j > 0$ for all $j \in S$ and $|\tilde{c}_j| \leq \mu'$ for all $j \notin S$. If moreover $|\tilde{c}_j| < \mu'$ for all $j \notin S$, the minimiser on $D \setminus R$ is unique and equals $\tilde{\beta}$ (extended by zeros).

Under the constant rule $\delta = 0$ and the last terms of (2)–(3) vanish.

Proof. A vector β with signed support (S, s) minimises the objective on $D \setminus R$ if and only if it satisfies the KKT conditions

$$X_{S,-R}^\top(y_{-R} - X_{S,-R}\beta_S) = \mu' s, \quad |X_{j,-R}^\top(y_{-R} - X_{S,-R}\beta_S)| \leq \mu' \quad (j \notin S), \quad s_j \beta_j > 0 \quad (j \in S). \quad (4)$$

The matrix of the linear system is $A' = X_{S,-R}^\top X_{S,-R} = A - U^\top U$, and by the matrix determinant lemma $\det A' = \det A \cdot \det(I - UA^{-1}U^\top) = \det A \cdot \det(I - H_R)$.

(a) If $I - H_R$ is singular then $X_{S,-R}$ is rank deficient, and by Lemma 2.4 no unique minimiser on $D \setminus R$ has support S .

(b) If $I - H_R$ is nonsingular, the Sherman–Morrison–Woodbury identity [10, 20, 24] gives $A'^{-1} = A^{-1} + A^{-1}U^\top(I - H_R)^{-1}UA^{-1}$, and the unique solution of the linear system is $\beta_S = A'^{-1}(X_S^\top y - U^\top y_R - \mu' s)$. Write $X_S^\top y - \mu s = A\hat{\beta}_S$ (the KKT equation on D) so that $X_S^\top y - U^\top y_R - \mu' s = A\hat{\beta}_S - U^\top y_R + \delta s$. Then

$$\begin{aligned} A'^{-1}(A\hat{\beta}_S - U^\top y_R) &= \hat{\beta}_S + A^{-1}U^\top[(I - H_R)^{-1}U\hat{\beta}_S - y_R - (I - H_R)^{-1}H_R y_R] \\ &= \hat{\beta}_S - A^{-1}U^\top(I - H_R)^{-1}(y_R - U\hat{\beta}_S), \end{aligned}$$

using $y_R + (I - H_R)^{-1}H_R y_R = (I - H_R)^{-1}y_R$; since $y_R - U\hat{\beta}_S = r_R$ and $A^{-1}U^\top = V_R^\top$, this is $\hat{\beta}_S - V_R^\top e_R$. The remaining term $\delta A'^{-1}s = \delta z$ by the same identity, which proves (2). For $j \notin S$, with $\beta_S = \hat{\beta}_S - V_R^\top e_R$ the full-length residual is $\tilde{r} = r + X_S V_R^\top e_R$, whose R -block is $r_R + H_R e_R = (I - H_R)^{-1}r_R = e_R$; hence

$$\begin{aligned} X_{j,-R}^\top \tilde{r} &= X_j^\top \tilde{r} - X_{R,j}^\top e_R = c_j + X_j^\top X_S A^{-1}U^\top e_R - X_{R,j}^\top e_R \\ &= c_j - (X_{R,j} - UA^{-1}X_S^\top X_j)^\top e_R = c_j - \tilde{X}_{R,j}^\top e_R, \end{aligned}$$

and the additional coefficient δz contributes $-\delta X_{j,-R}^\top X_{S,-R}z = -\delta(X_j^\top X_S - X_{R,j}^\top U)z$, which proves (3). The equivalence is now (4). If the inactive inequalities are strict, the equicorrelation set on $D \setminus R$ is S and $X_{S,-R}$ has full column rank, so the minimiser is unique [22]. $\square$

The test costs $O(k^3 + pk)$ per set R after the $O(nps_0 + s_0^3)$ precomputation of V , h , $\tilde{X}$ and c ; Section 6 measures its throughput against refits. Boundary cases ($s_j \beta_j = 0$ or $|\tilde{c}_j| = \mu'$) have probability zero for continuous designs and are counted, not hidden, in the validation.

The test decides whether (S, s) is preserved, which is not quite the same as deciding whether R is a witness. For the signed ANY target, failure of the test (or case (a)) identifies a witness in the sense of Definition 2.2 provided the minimiser on $D \setminus R$ is unique: for designs with a continuous distribution this holds for every R with probability one (the columns of X_{-R} are in general position, 22), which is how the exhaustive searches of Section 6 use the test; for integer data (Example 4.3) uniqueness is certified separately in exact arithmetic. For the other targets the flagged sets are refitted and the target is read from the new support.

Corollary 3.2 (Single removal; a stability margin). [proved here] Under the constant rule let $R = \{i\}$ and suppose $h_i < 1$ (equivalently, $X_{S,-i}$ has full column rank; $h_i = 1$ is the rank-deficient case of Proposition 3.1(a), in which (S, s) is not preserved). Put $e_i = r_i/(1 - h_i)$. Then

$$\tilde{\beta}_S = \hat{\beta}_S - v_i e_i, \quad \tilde{c}_j = c_j - \tilde{x}_{ij} e_i \quad (j \notin S),$$

so removing observation i preserves (S, s) if and only if $s_j(\hat{\beta}_j - v_{ij}e_i) > 0$ for $j \in S$ and $|c_j - \tilde{x}_{ij}e_i| < \mu$ for $j \notin S$ (boundary cases aside). With $m = \min_{j \in S} |\hat{\beta}_j|$ and $\gamma = \min_{j \notin S} (\mu - |c_j|)$ (if $S = \emptyset$, $m = +\infty$ and the

first term below is 0; if $S^c = \emptyset$, $\gamma = +\infty$ and the second term is 0; $\gamma = 0$ is a KKT boundary case and is excluded), define the instability index

$$\iota_i = |e_i| \max \left\{ \frac{\|v_i\|_\infty}{m}, \frac{\max_{j \notin S} |\tilde{x}_{ij}|}{\gamma} \right\} \quad \text{if } h_i < 1, \quad \iota_i = +\infty \quad \text{if } h_i = 1.$$

Then $\iota_i < 1$ implies that removing i preserves (S, s) , and $\max_i \iota_i < 1$ implies $f_{\text{ANY}}^{\text{signed}}(D) \geq 2$. Both statements are checkable in $O(np)$ time after the precomputation.

Proof. Specialise Proposition 3.1 to $k = 1$: $H_R = h_i$ and $I - H_R = 1 - h_i$ is nonsingular exactly when $h_i < 1$, since $\det A' = (1 - h_i) \det A$. The sufficient condition is the triangle inequality applied to (2)–(3); when $h_i = 1$ the removal does not preserve (S, s) by part (a), consistently with $\iota_i = +\infty$. The hypothesis is not vacuous: for $n = p = 2$, $X = I_2$, $y = (3, 3)^\top$, $\mu = 1$ the minimiser $\hat{\beta} = (2, 2)^\top$ is unique and $h_1 = h_2 = 1$. $\square$

Remark 3.3 (Relation to case-deletion diagnostics). Here $r = y - X\hat{\beta}$ is the residual of the Lasso fit on D , which differs from the least-squares residual of y on X_S by $\mu X_S A^{-1} s$, and $e_i = r_i / (1 - h_i)$ is the leave-one-out residual of the fixed-support fit. Under the constant rule, $v_i e_i = A^{-1} x_{iS} e_i$ is the DFBETA formula of Belsley et al. [3] for the regression of y on X_S with r_i in place of the least-squares residual (under rule P the extra term δz appears); $\tilde{c}_j - c_j = -\tilde{x}_{ij} e_i$ has no least-squares counterpart and is what decides entries. Cook’s distance [6] of the active-set regression is proportional to $h_i e_i^2$ and is used below as a target-agnostic heuristic. [classical]

Proposition 3.4 (Polynomial certificate for k removals, constant rule). [proved here] For a vector $a \geq 0$ let $T_k(a)$ be the sum of its k largest entries. Suppose $T_k(h) < 1$. Then for every R with $|R| \leq k$ the matrix $I - H_R$ is nonsingular and, with $\tilde{\beta}$ as in (2) and $\delta = 0$,

$$\begin{aligned} |\tilde{\beta}_j - \hat{\beta}_j| &\leq B_j(k) := T_k(|v_{\cdot j}| \odot |r|) + \sqrt{T_k(v_{\cdot j}^2) T_k(r^2)} \frac{T_k(h)}{1 - T_k(h)} \quad (j \in S), \\ |\tilde{c}_j - c_j| &\leq C_j(k) := T_k(|\tilde{x}_{\cdot j}| \odot |r|) + \sqrt{T_k(\tilde{x}_{\cdot j}^2) T_k(r^2)} \frac{T_k(h)}{1 - T_k(h)} \quad (j \notin S). \end{aligned}$$

Consequently, if $B_j(k) < |\hat{\beta}_j|$ for all $j \in S$ and $C_j(k) < \mu - |c_j|$ for all $j \notin S$, then no removal of at most k observations changes the signed support: $f_{\text{ANY}}^{\text{signed}}(D) \geq k + 1$. The certificate costs $O(nps_0 + pn \log n)$ (the T_k of $p + 1$ vectors of length n are read off after sorting each once; $O(np)$ with selection instead of sorting) and is monotone in k ; we write $k^*(D)$ for the largest certified k ($k^* = 0$ if none).

Proof. H_R is a principal submatrix of a projection, so $0 \preceq H_R \preceq I$ and $\lambda_{\max}(H_R) \leq \text{tr } H_R = \sum_{i \in R} h_i \leq T_k(h) < 1$; hence $I - H_R$ is nonsingular with $\|(I - H_R)^{-1}\|_2 \leq (1 - T_k(h))^{-1}$ and $\|e_R\|_2 \leq \|r_R\|_2 / (1 - T_k(h)) \leq \sqrt{T_k(r^2)} / (1 - T_k(h))$. From $(I - H_R)e_R = r_R$ we have $e_R = r_R + H_R e_R$, so

$$\begin{aligned} \tilde{\beta}_j - \hat{\beta}_j &= -v_{R,j}^\top e_R = -\sum_{i \in R} v_{ij} r_i - v_{R,j}^\top H_R e_R, \\ |v_{R,j}^\top H_R e_R| &\leq \|v_{R,j}\|_2 \|H_R\|_2 \|e_R\|_2 \leq \frac{\sqrt{T_k(v_{\cdot j}^2) T_k(r^2)} T_k(h)}{1 - T_k(h)}, \end{aligned}$$

and the first term is bounded by $T_k(|v_{\cdot j}| \odot |r|)$. The bound for $\tilde{c}_j$ is identical with $\tilde{x}_{\cdot j}$ in place of $v_{\cdot j}$. Under the hypotheses, every R with $|R| \leq k$ satisfies the conditions of Proposition 3.1(b) strictly, so the signed support is preserved. $\square$

The first term of $B_j(k)$ is the additive, “sum of single-deletion effects” approximation on which sorted-influence heuristics rest [4]; the second term controls the interaction between removed observations. Since $\sum_i h_i = s_0$, the hypothesis $T_k(h) < 1$ restricts the certificate to $k < n/s_0$ roughly, and the experiments show that it is loose (Section 6). Certified lower bounds on the number of removals that can overturn a least-squares conclusion, for far larger problems, are given by Rubinstein and Hopkins [19]; Proposition 3.4 is a crude bound of this kind for the Lasso signed support.

Remark 3.5 (Stability selection). [proved here] Stability selection [15] refits on subsamples of size $\lfloor n/2 \rfloor$ with a fixed penalty and reports the selection frequency Π_j of variable j . In the original paper the Lasso is $\|Y - X\beta\|_2^2 + \lambda \|\beta\|_1$ with λ fixed across subsamples, i.e. rule C (with $\mu = \lambda/2$); software that keeps a per-observation penalty fixed uses rule P. A subsample of size $\lfloor n/2 \rfloor$ is $D \setminus R$ with $|R| = \lceil n/2 \rceil$. Hence, if the Lasso minimiser on every subsample of size $\lfloor n/2 \rfloor$ is unique (with probability one for designs with

data set	support	exact coefficients	exact inactive c_j	marginal $x_1^\top y, x_2^\top y$
D	$\{1\}$	$\hat{\beta}_1 = \frac{3}{20}$	$c_2 = -\frac{1}{10}$	5, -1
$D \setminus R, R = \{3, 4\}$	$\{2\}$	$\hat{\beta}_2 = -\frac{5}{12}$	$c_1 = \frac{11}{4}$	4, -6
$D \setminus R', R' = \{2, 3, 4\}$	$\{1\}$	$\hat{\beta}_1 = \frac{1}{10}$	$c_2 = -\frac{17}{10}$	4, -2

Table 1: Example 4.3 in exact arithmetic ($\mu = \frac{7}{2}$, rule C). Coefficients solve the KKT system on the listed support; inactive correlations $c_j = X_j^\top (y - X\hat{\beta})$ satisfy $|c_j| < \mu$ strictly, which certifies the unique minimiser.

a continuous distribution), $f_{\text{LEAVE}(j)}(D) > \lceil n/2 \rceil$ under the rule used implies $\Pi_j = 1$; without uniqueness a subsample on which the solver drops j need not be a witness (Definition 2.2). The converse cannot be obtained from the definitions: $f_{\text{LEAVE}(j)} \leq \lceil n/2 \rceil$ produces a witness R of that size or smaller, but by Section 4 a superset of R of size exactly $\lceil n/2 \rceil$ need not be a witness. We did not search for an instance in which the converse actually fails.

4 Non-monotonicity

Proposition 4.1. *For each of the targets LEAVE, ENTER and ANY there exist data D and μ for which the target is not monotone, under both penalty rules.*

Example 4.2 ($p = 1$, by hand). [proved here] Let $x = (1, 1, 1)^\top$, $y = (2, -2, 2)^\top$, $\mu = \frac{3}{2}$. For $p = 1$ the minimiser is $\hat{\beta} = \text{sgn}(x^\top y)(|x^\top y| - \mu)_+ / \|x\|^2$, so the variable is selected iff $|x^\top y| > \mu$. On D , $x^\top y = 2 > \frac{3}{2}$: selected. On $D \setminus \{1\}$, $x^\top y = 0$: not selected under rule C ($\mu' = \frac{3}{2}$) and under rule P ($\mu' = 1$). On $D \setminus \{1, 2\}$, $x^\top y = 2$: selected under rule C ($2 > \frac{3}{2}$) and under rule P ($\mu'' = \frac{1}{2}$). Thus $\{1\}$ is a LEAVE witness and its superset $\{1, 2\}$ is not: $R = \{1\}$, $R' = \{1, 2\}$ is a non-monotone pair for LEAVE and for ANY. For ENTER take $x = (1, 1, 1, 1)^\top$, $y = (2, -2, 2, -2)^\top$, $\mu = \frac{3}{2}$: the variable is not selected on D ($x^\top y = 0$), is selected after removing observation 2 ($x^\top y = 2$, above $\mu' = \frac{3}{2}$ under rule C and $\mu' = \frac{9}{8}$ under rule P), and is not selected after removing $\{2, 3\}$ ($x^\top y = 0$); both removal sets are proper. Both examples are checked in exact arithmetic by the script of Appendix A (12/12 data-set/rule combinations).

The next example shows a mechanism specific to selection: a variable is excluded not because its own correlation with the response drops below the penalty, but because a competing variable absorbs it.

Example 4.3 ($p = 2$, competition; exact arithmetic). [computationally verified (exact arithmetic)] Let $n = 5$, $p = 2$, $\mu = \frac{7}{2}$ and

$$[X \mid y] = \begin{bmatrix} 2 & -1 & 2 \\ 0 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & -1 \\ -1 & 1 & 0 \end{bmatrix} \quad (\text{rows are observations; last column is } y),$$

with rule C, $R = \{3, 4\}$ and $R' = \{2, 3, 4\}$. Table 1 lists, for D , $D \setminus R$ and $D \setminus R'$, the signed support returned by the homotopy solver and the exact rational solution of the KKT system on that support, with all inequalities checked strictly in exact arithmetic (so the support is that of the unique minimiser). Variable 1 is selected on D , leaves on $D \setminus R$ while variable 2 enters, and returns on $D \setminus R'$. The exclusion is due to competition: on $D \setminus R$ the marginal correlation $|x_1^\top y| = 4$ exceeds $\mu = \frac{7}{2}$, so with $p = 1$ variable 1 would be selected, but the residual correlation after fitting variable 2 is $c_1 = \frac{11}{4}$, below μ . The instance was found by a seeded search over integer data in $\{-2, \dots, 2\}^{5 \times 2} \times \{-3, \dots, 3\}^5$ and rational μ (11 trials); all 26 removal sets with $|R| \leq n - 2$ (including $R = \emptyset$) are exactly verifiable (no KKT ties), and among them there are 3 non-monotone pairs for LEAVE(1).

By non-monotonicity, a minimal witness cannot be found by growing a set until the target holds and then pruning; exhaustive search by increasing size, or a heuristic followed by verification, are the honest options at small n .

5 Computational status

For $p = 1$, with $a_i = x_i y_i$ and $T = \sum_i a_i$, the variable is selected on $D \setminus R$ iff $|\sum_{i \notin R} a_i| > \mu |R|$. The problem is therefore about the sums of subfamilies of the numbers a_i , and the two directions of change behave very differently.

Theorem 5.1 ($p = 1$). [proved here] Consider instances with $p = 1$, rational data and rational $\mu > 0$, under either penalty rule.

- (a) DESELECTION-WITNESS: given (x, y, μ) with the variable selected on D , decide whether some nonempty proper $R \subsetneq [n]$ deselects it. This problem is NP-complete. Consequently computing f_{LEAVE} (equivalently the unsigned f_{ANY} , since for $p = 1$ every unsigned change is a deselection) is NP-hard, already for $p = 1$, hence for the general problem.
- (b) SELECTION-WITNESS: given (x, y, μ) with the variable not selected on D , the minimal $|R|$ such that it is selected on $D \setminus R$ can be computed in $O(n \log n)$ time.

Proof. (a) Membership in NP: given R , compute $\sum_{i \notin R} a_i$ and compare with $\mu_{|R|}$. Hardness: reduce from SUBSET SUM [9, 12]: given positive integers $b_1, \dots, b_m$ and a target $t \geq 1$, decide whether some subfamily sums to t ; without loss of generality $t < \sum_i b_i$ (otherwise the answer is immediate). Set $n = m + 1$, $x_i = 1$ for all i , $y_i = b_i$ for $i \leq m$, $y_{m+1} = -t$ and $\mu = \frac{1}{2}$. Then $a = (b_1, \dots, b_m, -t)$ and $T = \sum_i b_i - t \geq 1 > \mu \geq \mu_k$ for every k , so the variable is selected on D . For a nonempty proper R with $K = [n] \setminus R \neq \emptyset$, the variable is deselected on $D \setminus R$ iff $|\sum_{i \in K} a_i| \leq \mu_{|R|} \leq \frac{1}{2}$, iff $\sum_{i \in K} a_i = 0$ because the sum is an integer. If $m + 1 \notin K$ the sum is a nonempty sum of positive integers; hence $m + 1 \in K$ and $\sum_{i \in K \setminus \{m+1\}} b_i = t$, which is a solution of the SUBSET SUM instance, and conversely every solution K' gives the witness $R = [n] \setminus (K' \cup \{m + 1\})$, nonempty because $t < \sum_i b_i$ forces $K' \neq [m]$, and proper because $m + 1 \notin R$. The reduction is polynomial and works verbatim for rule P, since $\mu_k = \frac{1}{2}(n - k)/n < 1$.

(b) Suppose $|T| \leq \mu$. For k fixed, over all R with $|R| = k$ the sum $\sum_{i \notin R} a_i$ ranges over $[T - \sigma_k^+, T - \sigma_k^-]$ where σ_k^+ and σ_k^- are the sums of the k largest and of the k smallest a_i , and both endpoints are attained. Hence a witness of size k exists iff $T - \sigma_k^- > \mu_k$ or $T - \sigma_k^+ < -\mu_k$. After sorting the a_i once, all $\sigma_k^\pm$ are prefix sums and the smallest feasible $k \leq n - 1$ is found in $O(n)$. $\square$

Remark 5.2 (What the theorem does and does not say). (i) The hardness is inherited from SUBSET SUM and is therefore *weak*: for integer data a dynamic programme over partial sums computes f_{LEAVE} in time polynomial in n and in $\sum_i |a_i|$ (with the cardinality as a second state under rule P); strong NP-hardness (hardness for numbers polynomially bounded in n) is excluded for $p = 1$ by this pseudo-polynomial algorithm unless $P = NP$. (ii) For $p \geq 2$, LEAVE and unsigned ANY inherit the hardness of (a) through $p = 1$ instances padded with zero columns; Theorem 5.4 below adds ENTER, and Proposition 5.5 extends the pseudo-polynomial algorithm of (i) to every fixed p . (iii) The analogous question for the sign of a least-squares coefficient is the subject of Moitra and Rohatgi [16], Freund and Hopkins [8] and Rubinstein and Hopkins [19], and maximum feasible subsystems of linear relations are treated by Amaldi and Kann [1, 2]; we rely on no hardness statement from those papers and do not claim that the reductions here are new. [literature / not claimed]

Selection witnesses for $p \geq 2$. For $p = 1$ the selection event is that a sum leaves an interval, which is why Theorem 5.1(b) is easy. For $p = 2$ the residual correlation of an inactive variable has a kink: while a competing variable is inactive the correlation grows with the kept data, and once the competitor is active it absorbs the correlation. With integer data the kink leaves a window of width less than one, and the inactive variable enters exactly when a subset sum hits a target: the competition mechanism of Example 4.3 made exact. Full proofs of Lemma 5.3, Theorem 5.4 and Proposition 5.5 are in Appendix B.

Lemma 5.3 (Entry criterion for $p = 2$). [proved here] Let $p = 2$, $R \subsetneq [n]$, $K = [n] \setminus R$, $\mu' = \mu_{|R|}$, and suppose $G_{11} := \|X_{K,1}\|^2 > 0$. Put $G_{12} = X_{K,1}^\top X_{K,2}$, $q_l = X_{K,l}^\top y_K$ ($l = 1, 2$), $\beta_1^* = \text{sgn}(q_1)(|q_1| - \mu')_+/G_{11}$ and $\gamma = q_2 - G_{12}\beta_1^*$. (a) If $|\gamma| \leq \mu'$, then $(\beta_1^*, 0)$ minimises the Lasso objective on $D \setminus R$, and R is not a witness for ENTER(2). (b) If $|\gamma| > \mu'$, no minimiser on $D \setminus R$ has $\beta_2 = 0$; if the minimiser is unique, $2 \in S(D \setminus R)$.

Theorem 5.4 (Selection witnesses, fixed $p \geq 2$). [proved here] Fix $p \geq 2$ and either penalty rule. SELECTION-WITNESS: given rational (X, y, μ) with p columns, a unique minimiser on D and $j \notin S(D)$, decide whether some nonempty proper $R \subsetneq [n]$ is a witness for ENTER(j). This problem is NP-complete, also for integer data and already on instances on which the Lasso has a unique minimiser on $D \setminus R$ for every $R \subsetneq [n]$. Consequently computing $f_{\text{ENTER}(j)}$ is NP-hard for every fixed $p \geq 2$.

Sketch. For fixed p the certificate is R : the minimiser on $D \setminus R$ is found by enumerating the 3^p signed supports in exact arithmetic, and its uniqueness is decided by enumerating the vertices of the polytope of minimisers. Hardness is by reduction from SUBSET SUM with positive integers $b_1, \dots, b_m$, $B = \sum_i b_i$ and target $1 \leq t \leq B - 1$. Take $n = m + 1$, $p = 2$, item rows $x_i = (1, \rho)$ ($i \leq m$) and a gadget row $x_g = (u, \kappa u)$ with $y_g = 1/(2u)$, where $u = m + 1$, $\kappa = 1 + 1/(2B)$ and $\rho = 1 - 1/(8B\mu)$; under rule C, $y_i = b_i$ and $\mu = t + \frac{1}{2}$; under rule P, $y_i = b_i + \lambda$, $\lambda = t + \frac{1}{2}$ and $\mu = \lambda n$. If the gadget is kept and σ is the sum of the kept b_i , then

$\Delta := q_1 - \mu' = \sigma - t$ under both rules (under rule P the shift by λ cancels the dependence of μ' on $|K|$). By Lemma 5.3, for $\Delta \leq -1$ variable 1 is inactive and $0 < q_2 < \mu'$; for $\Delta \geq 1$ variable 1 is active and absorbs the correlation of variable 2 (because $u^2 > m$ and $\kappa - \rho \leq 1/B$); only $\Delta = 0$ lets variable 2 enter, with a unique minimiser. Without the gadget, $x_{i2} = \rho x_{i1}$ on K and variable 2 never enters. Hence R is a witness iff $g \notin R$ and the kept b_i sum to t , while on D ($\Delta = B - t \geq 1$) variable 2 is inactive. For $p \geq 3$ zero columns are appended. $\square$

Proposition 5.5 (Pseudo-polynomial algorithm for fixed p). [proved here] *Fix $p \geq 1$. For integer data with $|x_{ij}|, |y_i| \leq M$, rational $\mu > 0$, either rule and any target Π of Definition 2.2, $f_\Pi(D)$, and in particular whether $W_\Pi(D) = \emptyset$, can be computed with $O(n^2(2nM^2 + 1)^{p(p+3)/2})$ additions of integers of $O(\log(nM))$ bits, plus one test per reachable state whose cost depends only on p and is polynomial in the bit lengths of nM and μ .*

Idea. On $D \setminus R$ the objective is $\frac{1}{2}\beta^\top G_K \beta - q_K^\top \beta + \mu_{|R|}\|\beta\|_1$ plus a constant, with $G_K = X_K^\top X_K$ and $q_K = X_K^\top y_K$, so everything the target needs is a function of $(|K|, G_K, q_K)$, whose integer entries take at most $(n+1)(2nM^2 + 1)^{p(p+3)/2}$ values; a dynamic programme over the rows enumerates the reachable states.

Corollary 5.6 (The hardness is weak for every fixed p). [proved here] *For every fixed $p \geq 1$, DESELECTION-WITNESS (Theorem 5.1(a) padded with zero columns) and the existence of an unsigned ANY witness (which equals deselection of variable 1 in the padded instances, since zero columns never enter) are NP-complete, and so is SELECTION-WITNESS for every fixed $p \geq 2$. For integer data none of these problems is NP-hard in the strong sense unless $P = NP$; for integer data, strong NP-hardness therefore requires p to grow with the input.*

Proof. Hardness is Theorems 5.1(a) and 5.4; membership in NP is shown as in the proof of Theorem 5.4. Proposition 5.5 gives an algorithm polynomial in n , M and the bit length of μ , which a strongly NP-hard problem does not admit unless $P = NP$ [9]. $\square$

Remark 5.7 (Exhaustive check). [computationally verified (exact arithmetic)] `experiments/check_hardness.py` builds the instance of Theorem 5.4 from 156 SUBSET SUM instances (78 yes, 78 no: every target of three small vectors, and random ones with $m \leq 11$, so $n \leq 12$) and, for both rules, solves the Lasso in rational arithmetic on all 137,328 subsamples $D \setminus R$, certifying uniqueness by full rank of X_E [22]. A witness exists iff the source answer is yes in 312/312 (instance, rule) pairs; the set-by-set characterisation of the proof never fails; uniqueness is certified on every subsample, with no KKT boundary case; Proposition 3.1 (on every subsample) and the KKT-guarded solver (on 33,172 subsamples) agree with the exact solution; the dynamic programme of Proposition 5.5 reproduces $f_{\text{ENTER}(2)}$ in 312/312 pairs and agrees with exhaustive search in 145/145 (instance, target) pairs of random integer data with $p \in \{2, 3\}$. Three negative controls each violate one condition of the proof ($u = 1$, $\kappa = 2$, $\rho = 1$; 10 source instances each): the first two create spurious witnesses on purpose-built no-instances, the third leaves subsamples without certified uniqueness.

Conjecture 5.8 (Strong hardness for growing p). [conjectural] *With p part of the input, SELECTION-WITNESS and DESELECTION-WITNESS are NP-hard in the strong sense, already for data in $\{-1, 0, 1\}$ up to a common scale and a rational μ of polynomial size.*

A possible route, not carried out, uses d absorbing variables with integer slacks $\Delta_1, \dots, \Delta_d$: the target would enter only if $\Delta_l = 0$ for all l , an exact vector sum as in EXACT COVER BY 3-SETS [9]; what is missing is control of several active absorbers through the Gram matrix, including entries with the opposite sign.

6 Experiments

Protocol. All experiments use Gaussian designs with rows $x_i \sim N(0, \Sigma)$, $\Sigma_{jk} = \rho^{|j-k|}$, β^* with two nonzero entries $\pm b$, noise $N(0, 1)$, and $\mu = c\sqrt{2n \log p}$ (the universal threshold in sum form, scaled by c). Two families are used (their choice is discussed under “Pre-specification and changes” below): **A** (“strong signal”), $p = 4$, $b = 4$, $\rho = 0.0$, $c = 1.5$; **B** (“noisy”), $p = 8$, $b = 2$, $\rho = 0.5$, $c = 1.0$. The Lasso is solved by the homotopy algorithm [7, 17] as implemented in `sklearn.linear_model.lars_path`, whose output in E1–E5 satisfies the KKT equalities to 3.6×10^{-14} relative to $\max(1, \mu)$ and agrees in support with coordinate descent at tolerance 10^{-13} on all 104 refits checked (maximal coefficient difference 2×10^{-13}). Rule C is the default; rule P is reported where stated. Seeds are 20260930 (E0–E3, E5) and 20260931 (E4). Exhaustive searches enumerate removal sets by increasing size, decide the signed target with Proposition 3.1 and refit only the flagged sets when the target requires the new support; targeted searches are capped at $|R| \leq 5$ and the ANY search at $|R| \leq n - 2$ ($|R| \leq 6$ in E4), and a capped search that finds nothing is reported as “not found”, never as a value.

Criterion for “computational advantage” (fixed before the runs). A heuristic has computational advantage on a family if (i) it returns a witness of exactly minimal size in at least 90% of the instances whose exact fragility number is finite within the cap, and (ii) its median wall-clock cost is at most 10% of that of the exhaustive search on the same instances and machine.

Pre-specification and changes. The criterion above was fixed in the code (`run_fragility.py`) before any stored run, is recorded in `results/fragility.json`, and was never moved. Two choices were made after looking at data: (i) Families A and B were chosen after a pilot sweep (not included) over the penalty factor c at $n \leq 14$, in which almost all instances with $c \leq 1$ had $f = 1$; A is the most stable family found whose support is not always the true one, B a noisy one, and the fractions reported below describe these two families only. (ii) After a first complete run of E2, E3 and E5 the one-step greedy was corrected (its score became the predicted margin of the target instead of any break of the current pattern) and E2, E3 and E5 were rerun. All scripts of this version were run with the final library in one session. *Timing.* The exhaustive-search timer starts after the initial fit and the $O(nps_0)$ precomputation, whereas the heuristic timers include the initial fit (about one refit, $670 \mu s$ at $n = 14$); subtracting from every heuristic its mean time per refit step (which includes one scoring step and so favours the heuristic) leaves the smallest median cost ratio over all combinations at 0.25 (raw: 0.44), so no verdict changes.

6.1 E1: validation of Proposition 3.1 and of the certificates

On four (n, p) cells ($b = 2, \rho = 0.3, c = 1$), with 15 instances per cell for the comparison with refits and 30 fresh instances per cell for the certificates, Proposition 3.1 was compared with a refit for every single removal and for random sets of 2, 3 and 5 observations, under both rules. Agreement was 2760/2760 for single removals and 7200/7200 for sets, and whenever the signed support was preserved the closed-form coefficients (2) matched the refit to 4×10^{-15} (single) and 6×10^{-15} (sets); the per-cell counts are in `results/validation.md`. The sufficient condition of Corollary 3.2 certified 1216 of the 2114 single removals that are in fact stable (58%), and never a removal that is not. On $n = 14, p = 6$, all 1001 subsets of size 4 were tested at $1.8 \mu s$ per subset against $670 \mu s$ per refit, a factor of 369. Per-cell results for Proposition 3.4 are in `results/validation.md` and are summarised with E2.

6.2 E2: exact fragility numbers for $n \leq 14$

For 60 instances per cell we computed the exact signed fragility number for ANY under both rules, the type of change realised by the minimal witnesses, and the targeted fragility numbers for the weakest selected variable (smallest $|\hat{\beta}_j|$) and for the strongest inactive variable (largest $|c_j|$); the per-cell table is `results/fragility.md`, pooled values follow. In family B the selected set changes after a single removal in 92% of the 240 instances (maximal $f = 3$); in family A in 52% (maximal $f = 4$). Family B had 8 instances with an empty support, 2 of them without a witness within $|R| \leq n - 2$ (counted as not found); signed and unsigned ANY differed in 0 (A) and 0 (B) instances. Among 20 of the (up to 50) collected minimal witnesses per instance, the change is a single variable leaving in 51% (A) and 54% (B) of the cases, a single variable entering in 43% and 25%, and a simultaneous leave and enter in 6% and 20%; sign-only changes occurred 0 and 0 times. Targeted searches were not concluded within the cap in 12/240 (A, LEAVE) and 31/237 (A, ENTER) instances. The certificate of Proposition 3.4 gave $k^* \geq 1$ in 30% of the 115 family-A instances with $f \geq 2$ (mean gap 1.2).

6.3 E3: heuristics against the exact minimum

Three heuristics were run on the same instances (rule C; E3 uses the unsigned ANY target, which coincided with the signed one in every instance): *one-step greedy*, which at every step removes the observation whose exact single-removal effect (Corollary 3.2 on the current fit) brings the target closest, refits, and repeats; *sorted scores*, in the style of the approximate maximum influence perturbation [4], which scores every observation once on the full data by its exact single-removal effect on each margin, predicts the number of removals k_{pred} from cumulative sums, and then removes observations in that fixed order with a refit after each until the target holds; and *Cook’s distance*, target-agnostic. Table 2 pools the results over $n \leq 14$ against the predefined criterion. 0 of the 18 (family, target, heuristic) combinations met it. The one-step greedy is the most accurate: it returns a minimal witness in 97%/100% of the instances for ANY (families A/B), 95%/98% for LEAVE and 86%/88% for ENTER (maximal excess over the minimum in family A: 4). These are point estimates on 206–240 instances with standard errors of 0.4–3.1 points: the one-step greedy meets the accuracy part of the criterion for ANY and LEAVE, while for ENTER its shortfall (86%/88% against 90%) is 1.7 and 1.1 standard errors below the threshold and should not be read as a settled failure. The

fam.	target	heuristic	n	exact $\pm$ s.e.	n ($f \geq 2$)	exact ($f \geq 2$)	cost ratio	adj. cost ratio	advantage
A	any	one-step greedy	240	97% $\pm$ 1.2	115	93%	1.3	0.80	no
A	any	sorted scores (AMIP-style)	240	82% $\pm$ 2.5	115	63%	1.2	0.77	no
A	any	Cook's distance	240	66% $\pm$ 3.1	115	57%	1.2	0.73	no
A	leave	one-step greedy	228	95% $\pm$ 1.4	167	93%	0.48	0.30	no
A	leave	sorted scores (AMIP-style)	228	75% $\pm$ 2.9	167	65%	0.44	0.28	no
A	leave	Cook's distance	228	67% $\pm$ 3.1	167	66%	0.46	0.30	no
A	enter	one-step greedy	206	86% $\pm$ 2.4	133	78%	0.75	0.47	no
A	enter	sorted scores (AMIP-style)	206	71% $\pm$ 3.1	133	56%	0.71	0.46	no
A	enter	Cook's distance	206	27% $\pm$ 3.1	133	11%	1.0	0.73	no
B	any	one-step greedy	238	100% $\pm$ 0.0	17	100%	0.77	0.40	no
B	any	sorted scores (AMIP-style)	238	99.6% $\pm$ 0.4	17	94%	0.78	0.40	no
B	any	Cook's distance	238	87% $\pm$ 2.2	17	76%	0.63	0.34	no
B	leave	one-step greedy	232	98% $\pm$ 0.9	48	96%	0.56	0.28	no
B	leave	sorted scores (AMIP-style)	232	95% $\pm$ 1.5	48	79%	0.53	0.27	no
B	leave	Cook's distance	232	78% $\pm$ 2.7	48	65%	0.50	0.26	no
B	enter	one-step greedy	209	88% $\pm$ 2.3	69	80%	0.46	0.26	no
B	enter	sorted scores (AMIP-style)	209	85% $\pm$ 2.5	69	72%	0.46	0.25	no
B	enter	Cook's distance	209	23% $\pm$ 2.9	69	1%	0.74	0.51	no

Table 2: E3. Heuristics against the exact minimum, pooled over $n \in \{8, 10, 12, 14\}$, rule C. “Exact” is the fraction of instances (with finite exact f within the cap) in which the heuristic’s witness has minimal size; “s.e.” is the binomial standard error in percentage points; “cost ratio” is the median of heuristic time over exhaustive-search time, and “adj.” subtracts the heuristic’s mean time per refit step (at least one refit) from the heuristic time; interquartile ranges are in `results/fragility.md`. The verdict applies the criterion fixed before the run (exact $\geq 90\%$ and median cost ratio $\leq 10\%$).

sorted-scores heuristic is less accurate (82%/99.6%, 75%/95%, 71%/85%) and Cook’s distance is the worst for the targeted changes. No heuristic meets the cost part: at these sizes none is cheaper than one tenth of the exhaustive search that uses Proposition 3.1 (smallest median cost ratio over all combinations 0.44, or 0.25 with the adjustment described under *Timing*), per combination in Table 2. The linear prediction k_{pred} of the sorted-scores heuristic equals the exact fragility number in 70%/99% of the ANY instances (A/B) and underestimates it in 0%/0%.

6.4 E4: scaling with n

With the closed-form test the exhaustive search for signed ANY extends to $n = 34$ at $|R| \leq 6$ (Figure 1a,b; per-cell table in `results/scaling.md`). In family A the censored mean fragility number (instances above the cap excluded, which biases the mean downwards as n grows; medians are in `results/scaling.md`) rises slowly with n (1.43 at $n = 10$, 4.00 at $n = 34$, where 1 of 20 instances lie above the cap), while in family B it stays close to one (75% of the instances have $f = 1$ at $n = 34$). Over the 233 family-A instances with finite f the one-step greedy returned the exact minimum in 88% of the cases (83% of the 174 with $f \geq 2$, maximal excess 6). Its cost relative to the exhaustive search depends on the family and on n , and wall-clock ratios vary between complete runs of the same code (cell medians changed by up to a factor 2.3, in family A at $n = 22$, over three runs), so ranges over the runs are quoted (two significant digits). In family B at every $n \leq 34$ (median ratios 8.1–11) and in family A at $n \leq 18$ (4.3–11) the greedy is slower than the exhaustive search: when f is small, one batch of n closed-form tests (median search time at most 0.13 ms in family B) beats the two or more fits of the greedy (1.2–1.5 ms). In family A the ratio is close to one at $n = 22$ –26 (0.60–1.4), below one at $n = 30$ (0.58–0.68) and below the 10% threshold only at $n = 34$ (0.055–0.070), where the greedy was exact in 17 of the 19 instances with finite f (89%, standard error 7 points); 20 instances do not decide the accuracy part of the criterion at that size, and exactness is not monotone in n (77, 93, 82 and 89% at $n = 22, 26, 30, 34$). Beyond exhaustive reach, Figure 1c,d brackets the fragility number between the certificate of Proposition 3.4 ($k^* + 1$) and the greedy upper bound g : at $n = 400$ in family A the median greedy witness has 10.5 observations (mean $g/n \approx 0.028$), the median certified lower bound is $k^* + 1$ with $k^* = 5$, and the mean ratio $(k^* + 1)/g$ is 0.57; the bracket is wide, and the observed decrease of g/n with n is an empirical statement about these families only.

6.5 E5: frequency of non-monotone pairs

For every instance of E2 with a finite targeted fragility number, all one-element supersets $R \cup \{i\}$ of 20 of the (up to 50) collected minimal witnesses R were refitted. For LEAVE of the weakest variable, the variable had re-entered the support in 7.1% of the 6960 superset pairs of family A (at least one re-entry in 51% of the 228 instances) and in 10.5% of 8092 pairs of family B (72% of 232 instances). For ANY, the original support was restored by adding one observation to a minimal witness in 14.9% (A) and 9.3% (B) of the pairs (per-cell values in `results/fragility.md`). Non-monotonicity is thus a regular feature of minimal witnesses

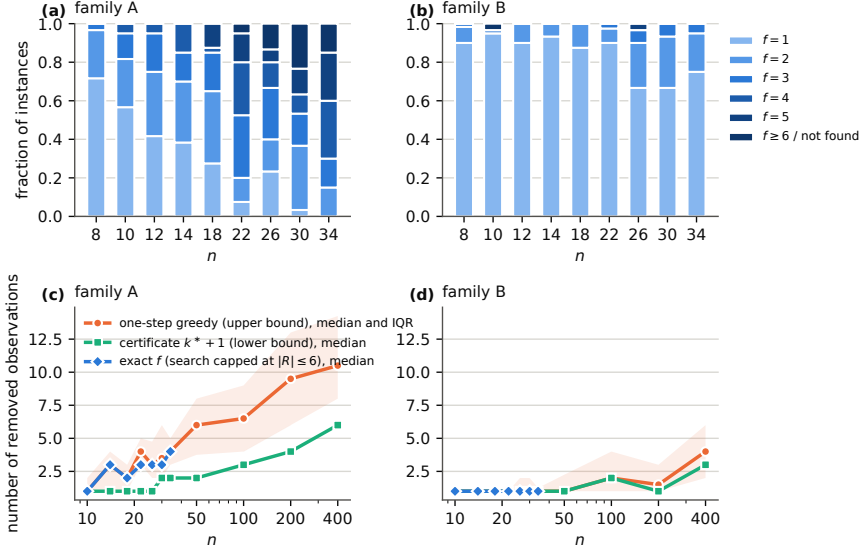


Figure 1: (a,b) Distribution of the exact signed fragility number for ANY (rule C) against n , families A and B, from E2 ($n \leq 14$) and E4 ($n \geq 18$; search capped at $|R| \leq 6$, the top segment collects $f \geq 6$ and not found). (c,d) Bracketing at large n (rule C): median one-step greedy witness size (upper bound, with interquartile range), median certified lower bound $k^* + 1$ from Proposition 3.4, and exact medians from E4 where available.

on these designs.

7 What can be claimed

claim	status
A unique Lasso minimiser has X_S of full column rank (Lemma 2.4); uniqueness under full rank of X_E (22)	classical / literature
Exact closed-form test of whether removing R preserves the signed support, both rules (Prop. 3.1); agreement with refits in 2760/2760 single and 7200/7200 multiple removals (E1); identifies signed-ANY witnesses when the minimiser on $D \setminus R$ is unique	proved here; verified
Single removal with $h_i < 1$ (rule C) is a DFBETA-type formula with the Lasso residual; $O(np)$ certificate for $f \geq 2$ (Cor. 3.2)	proved here
Polynomial certificate for $f \geq k + 1$ (Prop. 3.4); certifies $k^* \geq 1$ in 30% of family-A instances with $f \geq 2$, $n \leq 14$: loose (E1, E2, E4)	proved here; verified
Witness families are not monotone: hand-checkable example with $p = 1$ (Ex. 4.2); competition example with $p = 2$, exact rational KKT certificate (Ex. 4.3)	proved here; verified (exact arithmetic)
$f_{\text{LEAVE}(j)} > \lceil n/2 \rceil$ implies stability-selection frequency 1 if the minimisers on all subsamples of size $\lceil n/2 \rceil$ are unique; converse not implied (Rem. 3.5)	proved here
Existence of a deselection witness is NP-complete for $p = 1$; fragility number NP-hard in general; selection witnesses for $p = 1$ in $O(n \log n)$ (Thm. 5.1)	proved here
Existence of a selection witness is NP-complete for every fixed $p \geq 2$, both rules, with unique minimisers on all subsamples (Lemma 5.3, Thm. 5.4); pseudo-polynomial algorithm for every fixed p and target (Prop. 5.5), so these problems are only weakly NP-complete for fixed p (Cor. 5.6); reduction checked exhaustively, 312/312 pairs (Rem. 5.7)	proved here; verified (exact arithmetic)
Strong NP-hardness with p part of the input (Conj. 5.8)	conjectural
Single-removal fragility ($f = 1$) in 92% (B) and 52% (A) of instances with $n \leq 14$; minimal witnesses mostly single events (E2); non-monotone pairs among one-element supersets of minimal witnesses in a measurable fraction of cases (E5)	verified, these two families only
No heuristic meets the predefined advantage criterion at $n \leq 14$ (all fail the cost part; the one-step greedy's shortfall for ENTER is within two s.e.); the one-step greedy is the most accurate; it is cheaper than the exhaustive search in all three runs only in family A at $n \geq 30$, and below the cost threshold only at $n = 34$, with 17/19 exact (E3, E4)	verified, these two families only
Rates of f/n with n ; behaviour on real data; novelty of the algebra or of the reductions	not claimed

Table 3: Claims and their status; “verified” means reproduced by the scripts of Appendix A with the fixed seeds, specific to the two synthetic families and not a proof.

Limitations. The exact test presupposes a unique minimiser and full column rank on the support; ties are measure-zero for continuous designs but real in integer data, and the code counts them rather than resolving

them. For the signed ANY target the exhaustive search records a non-preserved set as a witness without refitting, which presupposes uniqueness on $D \setminus R$: almost sure for the Gaussian designs used here and certified exactly for the integer examples, but not guaranteed for integer data in general. The homotopy solver mishandles exact ties of correlations (as in the instance in the proof of Corollary 3.2); every fit is therefore KKT-checked with a coordinate-descent fallback. The fallback was triggered only in the integer-data search of E0 (91 of 950 fits, all exact ties of $|x_j^\top y|$ at entry; the examples are identical with and without it) and never in E1–E5 (0 of 144,264 fits). The certificate of Proposition 3.4 is loose and restricted to $k < n/s_0$. The hardness statements are weak for every fixed p (Corollary 5.6), rest on specially structured data, and say nothing about typical instances, on which exhaustive search was fast. The experiments use two synthetic families chosen after a pilot sweep, one solver and a cap on $|R|$; the fragility numbers are exact under that cap and estimate nothing beyond the families, and the cost ratios depend on the implementation (a vectorised test against a Python loop of refits) and on machine load.

Next steps for the research line. (a) Prove or refute Conjecture 5.8, starting from the multi-absorber gadget sketched after it; for fixed p , improve the exponent $p(p+3)/2$ of Proposition 5.5. (b) Tighten Proposition 3.4 (sign structure of the interaction term, or a small semidefinite relaxation). (c) Data-driven penalties (cross-validation), for which the witness must account for the change of μ . (d) The elastic net and the square-root Lasso, whose KKT systems are also linear on a fixed signed support.

A Reproducibility

The library `experiments/lasso_fragility.py` and the scripts `exact_examples.py` (E0), `run_validation.py` (E1), `run_fragility.py` (E2, E3, E5), `run_scaling.py` (E4), `make_figures.py` and `make_numbers.py` (which turns `results/*.json` into the macros of this document, so that no number in the text is typed; the cost ranges of E4 also use the ratios of two earlier complete runs archived by `snapshot_cost_ratios.py`) are described in the README. The reference run (Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, scikit-learn 1.9.1 [18]; seeds 20260930 and 20260931) took 9/50/44 s for E1/E2–E3–E5/E4, 1.7 minutes in all, on one core of a shared four-core machine.

Exhaustive check of Section 5. `experiments/check_hardness.py` (a copy of `theory/check_hardness.py` that also writes `results/hardness.json`; seed 20261003, 38 s) solves every subsample in rational arithmetic (`fractions.Fraction`); its log is frozen in `results/check_hardness_output.txt` with SHA-256 prefix 2049f31ebeeec189b, which `make_numbers.py` checks before writing the macros of Remark 5.7.

Solver. `lars_path(X, y, method="lasso", alpha_min= $\mu/n'$ )` on the reduced data with n' rows; the last column of the path is the minimiser of (1) at penalty μ ; supports are read off its exact zeros (threshold 10^{-10}). Every fit is KKT-checked (relative tolerance 10^{-8}) and, if the check fails, recomputed by coordinate descent and checked again; this happens on exact ties of correlations (`check_b1_counterexample.py`), in the reference run in 91 of the 950 fits of E0 and in none of E1–E5 (counters in the field `meta.kkt_guard` of each `results/*.json`).

Exact test and search. For a batch of subsets of equal size k : H_R by one batched matrix product, e_R by a batched solve, $\det(I - H_R) \leq 10^{-12}$ treated as rank deficient (a change), conditions of Proposition 3.1(b) with strict inequalities; sizes $k = 1, 2, \dots$ in order, refits only for flagged subsets of the targeted searches, up to 50 minimal witnesses collected. Heuristics and exact checks: see the docstrings.

B Proofs for Section 5

Proof of Lemma 5.3. On $\{\beta_2 = 0\}$ the objective is $\frac{1}{2}\|y_K - X_{K,1}\beta_1\|^2 + \mu'|\beta_1|$ up to a constant, strictly convex because $G_{11} > 0$, with unique minimiser β_1^* (the formula of Example 4.2). At $(\beta_1^*, 0)$ the KKT condition for coordinate 1 is the optimality condition of this restricted problem, and the condition for coordinate 2 reads $|q_2 - G_{12}\beta_1^*| \leq \mu'$. (a) The KKT conditions are sufficient for the convex objective. If the minimiser is unique it is $(\beta_1^*, 0)$ and $2 \notin S(D \setminus R)$; if it is not unique, R is not a witness by Definition 2.2. (b) A minimiser $(\beta_1, 0)$ would minimise the restricted problem, so $\beta_1 = \beta_1^*$, and the necessary KKT conditions would give $|\gamma| \leq \mu'$. $\square$

Proof of Theorem 5.4. Membership in NP (fixed p). The certificate is R . By Lemma 2.4 a unique minimiser on $D \setminus R$ has a support S with $X_{K,S}$ of full column rank, so a candidate is found by enumerating the 3^p signed

supports, solving each KKT system $X_{K,S}^\top(y_K - X_{K,S}\beta_S) = \mu's$ in exact rational arithmetic (polynomial bit length by Cramer's rule) and checking the sign and inactive inequalities; if no candidate passes, the minimiser is not unique. Given a minimiser $\hat{\beta}$, every minimiser has the same fit and the same ℓ_1 norm [22], so the set of minimisers is the polytope $\mathcal{M} = \{\beta : X_K\beta = X_K\hat{\beta}, \|\beta\|_1 \leq \|\hat{\beta}\|_1\}$, and since $X_K v = 0$ iff $X_K^\top X_K v = 0$ the fit constraint is a system of p linear equations. If $\mathcal{M}$ contains a segment, some closed orthant contains a piece of it, and the intersection of $\mathcal{M}$ with that orthant has a vertex other than $\hat{\beta}$. Hence $\mathcal{M} = \{\hat{\beta}\}$ iff no such intersection has a vertex other than $\hat{\beta}$, which is decided by enumerating, in each of the 2^p closed orthants, the vertices cut out by the p sign constraints, the p equations and the norm constraint (linear in the orthant), in time polynomial for fixed p . Finally test $\hat{\beta}_j \neq 0$.

Reduction for $p = 2, j = 2$. Let $b_1, \dots, b_m$ be positive integers, $B = \sum_i b_i$ and t the target. If $t \leq 0$ or $t \geq B$ the answer is immediate, and we output the instance below for the fixed yes-instance $((1, 1), 1)$ or the fixed no-instance $((2, 2), 1)$; otherwise $1 \leq t \leq B - 1$. Put $n = m + 1$, $u = m + 1$, $\eta = 1/(2B)$, $\kappa = 1 + \eta$; under rule C, $\mu = t + \frac{1}{2}$ and $y_i = b_i$; under rule P, $\lambda = t + \frac{1}{2}$, $\mu = \lambda n$ and $y_i = b_i + \lambda$ ($i \leq m$); then $\varepsilon = \eta/(4\mu)$ and $\rho = 1 - \varepsilon$. The rows are $x_i = (1, \rho)$ with response y_i for $i \leq m$, and $x_g = (u, \kappa u)$ with $y_g = 1/(2u)$ for the gadget $g = m + 1$. All numbers are rationals with numerators and denominators bounded by a polynomial in m, B and t . We use $\kappa - \rho = \eta + \varepsilon \in (0, 1/B]$ (as $\varepsilon \leq \eta$), $\rho > \frac{1}{2}$, $u^2 > m$, $\mu' \leq \mu$, and $\mu' \geq \frac{3}{2}$ for every R (rule C: $\mu' = t + \frac{1}{2}$; rule P: $\mu' = \lambda|K| \geq \lambda$). Let $R \subseteq [n]$, $K = [n] \setminus R$, $I = K \cap [m]$, $N = |I|$, $\sigma = \sum_{i \in I} b_i$ and $W = \sum_{i \in I} y_i$; we use the notation of Lemma 5.3.

Case $g \notin K$ ($N \geq 1$). Then $G_{11} = N$, $G_{12} = \rho N$, $q_1 = W$, $q_2 = \rho W$ and $\gamma = \rho(W - \text{sgn}(W)(|W| - \mu')_+)$, so $|\gamma| = \rho \min(|W|, \mu') < \mu'$: not a witness, by Lemma 5.3(a).

Case $g \in K$. Then $G_{11} = u^2 + N$, $G_{12} = \kappa u^2 + \rho N$, $q_1 = \frac{1}{2} + W > 0$ and $q_2 = \frac{\kappa}{2} + \rho W > 0$. Put $\Delta = q_1 - \mu'$. Under rule C, $\Delta = \sigma - t$. Under rule P, $|K| = N + 1$, $\mu' = \lambda(N + 1)$ and $W = \sigma + \lambda N$, so again $\Delta = \sigma - t$; thus $\Delta \in \mathbb{Z}$ under both rules. Direct computation gives

$$q_2 - \mu' = \frac{1}{2}(\kappa - \rho) + \rho\Delta - (1 - \rho)\mu', \quad \frac{G_{12}}{G_{11}} - \rho = \frac{(\kappa - \rho)u^2}{u^2 + N}, \quad (5)$$

with $0 < (1 - \rho)\mu' = \varepsilon\mu' \leq \varepsilon\mu = \eta/4$, and $\beta_1^* = \Delta_+/G_{11}$ because $q_1 > 0$. (i) $\Delta \leq -1$: $\gamma = q_2 \in (0, \mu')$, because $q_2 - \mu' \leq \frac{1}{2}(\kappa - \rho) - \rho < 0$; no entry by Lemma 5.3(a). (ii) $\Delta = 0$: $\gamma = q_2$ and $q_2 - \mu' \geq \frac{1}{2}(\eta + \varepsilon) - \frac{1}{4}\eta > 0$. Here $\sigma = t \geq 1$, so $N \geq 1$; the rows $(1, \rho)$ and $(u, \kappa u)$ are linearly independent ($\kappa \neq \rho$), so X_K has rank 2, the objective is strictly convex and the minimiser unique, and $2 \in S(D \setminus R)$ by Lemma 5.3(b). (iii) $\Delta \geq 1$: $\gamma = q_2 - G_{12}\Delta/G_{11}$ and, by (5),

$$\gamma - \mu' = (\kappa - \rho)\left(\frac{1}{2} - \frac{\Delta u^2}{u^2 + N}\right) - (1 - \rho)\mu' < 0, \quad \gamma + \mu' > (1 + \rho)\mu' - (\kappa - \rho)\Delta \geq \frac{3}{2} - \frac{B - t}{B} > 0,$$

using $u^2 > m \geq N$ (so $\Delta u^2/(u^2 + N) > \frac{1}{2}$), $\Delta \leq B - t$ and $\kappa - \rho \leq 1/B$; no entry by Lemma 5.3(a). Hence R is a witness for ENTER(2) if and only if $g \notin R$ and $\sum_{i \in [m] \setminus R} b_i = t$.

Full data and equivalence. For $K = [n]$, $\Delta = B - t \geq 1$; by (iii) $(\beta_1^*, 0)$ with $\beta_1^* > 0$ is a minimiser with $|\gamma| < \mu$ (under rule P, $\mu' = \mu$ for $R = \emptyset$), unique because X has rank 2, so $S(D) = \{1\}$ and $2 \notin S(D)$. If $\sum_{i \in I} b_i = t$ for some $I \subseteq [m]$, then $\emptyset \neq I \neq [m]$ because $1 \leq t \leq B - 1$, and $R = [m] \setminus I$ is a nonempty proper witness; conversely, a witness yields such an I .

Uniqueness on every subsample. If $g \in K$ and $N \geq 1$, X_K has rank 2. If $K = \{g\}$ (respectively $g \notin K$), all rows of X_K are multiples of $(1, \kappa)$ (respectively $(1, \rho)$); all minimisers share the fit, hence the value of $\beta_1 + \kappa\beta_2$ (respectively $\beta_1 + \rho\beta_2$), and because $\kappa > 1$ (respectively $\rho < 1$) the ℓ_1 norm at a fixed value is minimised only at $\beta_1 = 0$ (respectively $\beta_2 = 0$), so the minimiser is unique.

Integer data and $p \geq 3$. For $\alpha, \alpha' > 0$, replacing (X, y, μ) by $(\alpha X, \alpha' y, \alpha\alpha'\mu)$ and substituting $\beta = (\alpha'/\alpha)\beta'$ multiplies the objective by α'^2 under both rules, so supports, uniqueness and witnesses are unchanged; clearing denominators gives integer data bounded by a polynomial in m, B and t . For fixed $p \geq 3$ append $p - 2$ zero columns: every minimiser vanishes on them, which leaves minimisers, uniqueness and supports unchanged. $\square$

Proof of Proposition 5.5. On $D \setminus R$ the objective equals $\frac{1}{2}\beta^\top G_K \beta - q_K^\top \beta + \mu'\|\beta\|_1$ plus a constant, with $\mu' = \mu_{|R|}$ a function of $|K|$. Hence the set of minimisers, its uniqueness, the signed support and the truth of Π are functions of the state $(|K|, G_K, q_K)$; for uniqueness, $X_K v = 0$ iff $G_K v = 0$, so the polytope $\mathcal{M}$ of the proof of Theorem 5.4 is $\{\beta : G_K(\beta - \hat{\beta}) = 0, \|\beta\|_1 \leq \|\hat{\beta}\|_1\}$. The entries of G_K and q_K are integers in $[-nM^2, nM^2]$, so there are at most $(n + 1)(2nM^2 + 1)^{p(p+1)/2+p}$ states. Processing the rows one at a time, the states reachable by subsets of the first i rows are obtained from those of the first $i - 1$ rows by adding row i or not, at cost $O(p^2)$ per state and row, which gives the stated count for fixed p . Finally test Π , with uniqueness, on every reachable state with $1 \leq |K| \leq n - 1$ (enumeration of the 3^p signed supports and the vertex enumeration of Theorem 5.4, on $O(p)$ constraints in p variables), and return n minus the largest $|K|$ that passes ($f_\Pi = \infty$ if none passes). $\square$

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